House Price Prediction

Model Comparison & Performance Optimization

California Housing Dataset — Feature Engineering, Model Optimization & Performance Comparison
AI & Machine Learning Internship — Task 2
Maincrafts Technology

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Internship Domain:	Artificial Intelligence & Machine Learning
Task:	Task 2 — Feature Engineering, Model Optimization & Performance Comparison

1. Project Overview

Objective

The objective of Task 2 is to move beyond a single baseline model and build a complete model comparison workflow. This includes preparing the data correctly through feature scaling, training multiple regression algorithms, evaluating each with standard metrics, and selecting the best-performing model with clear, evidence-based justification.

Dataset Description

The California Housing Dataset was used, containing real housing-related data across California districts. The target variable is Median House Value, while the input features include Median Income, House Age, Average Rooms, Average Bedrooms, Population, and location-based attributes (latitude and longitude).

Libraries Used

- Python — core programming language
- Pandas — data loading and manipulation
- NumPy — numerical operations
- Matplotlib / Seaborn — data visualization
- Scikit-learn — preprocessing, model training, and evaluation

Workflow

- Loading the California Housing dataset into a structured DataFrame
- Exploratory Data Analysis — inspecting nulls, ranges, and feature correlations
- Feature Scaling using StandardScaler, applied before model training
- Train-Test Split — dividing data into training and unseen test sets (80/20)
- Model Training — fitting Linear Regression, Ridge Regression, and Decision Tree Regressor
- Model Evaluation — computing MAE, RMSE, and R^2 Score for each model
- Model Comparison — tabulating results and selecting the best-performing model

2. Models & Results

Linear Regression

Linear Regression fits a straight-line relationship between the input features and the target variable, minimizing the overall prediction error. It served as the baseline model for this comparison.

Ridge Regression

Ridge Regression extends Linear Regression by adding a penalty that discourages any single feature's weight from growing too large. This keeps the model from over-relying on one feature and encourages a more balanced distribution of influence across all features, based on their actual importance rather than coincidental patterns in the training data.

Decision Tree Regressor

Decision Tree Regressor splits the data into branches based on feature thresholds, allowing it to capture non-linear relationships that a straight-line model cannot represent. A maximum depth of 5 was used to prevent the tree from overfitting to the training data.

Why do Linear Regression and Ridge look so similar?

Ridge Regression is simply Linear Regression with a built-in “safety brake” that discourages any one feature from getting an oversized weight. That brake only makes a visible difference when the plain Linear Regression model was leaning too heavily on a risky or coincidental pattern in the training data. In this dataset, Linear Regression was not doing that — its feature weights were already reasonably behaved. So Ridge's penalty had nothing significant to correct, which is exactly why both models produced almost identical MAE, RMSE, and R^2 values. This is itself a useful finding: it tells us overfitting from oversized coefficients was not a real problem in the baseline model.

Model Performance Comparison

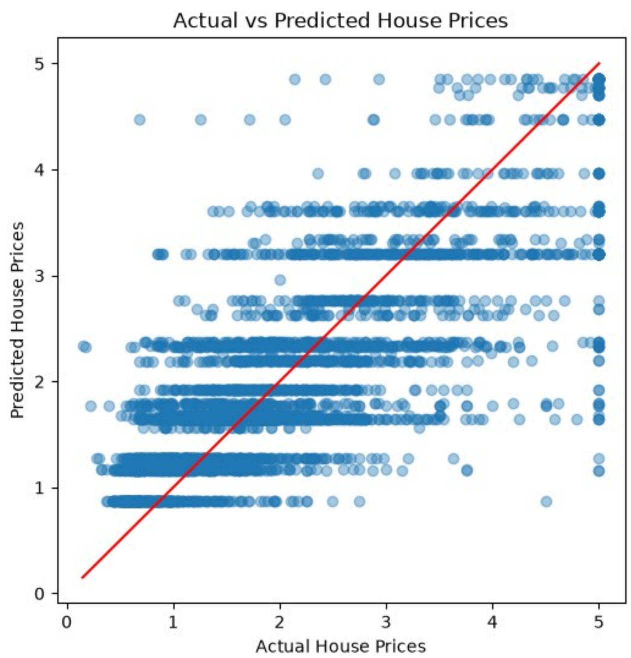
Model	MAE	RMSE	R^2 Score
Linear Regression	0.533	0.746	0.576
Ridge Regression	0.533	0.746	0.576
Decision Tree Regressor ★	**0.522**	**0.724**	**0.600**

Interpretation

- Lower MAE and RMSE indicate better prediction accuracy.
- Higher R^2 Score indicates better explanatory power of the model.
- Decision Tree Regressor achieved the best overall performance across all three metrics.

3. Visualization & Conclusion

Actual vs Predicted House Prices



Predicted values cluster around the ideal reference line most closely in the mid-price range, while a visible flattening near the upper bound reflects the dataset's known price cap at $500,000 — a limitation of the data rather than the model.

Key Findings

- Feature scaling was essential for Linear and Ridge Regression, as it placed all features on a common scale and improved training stability.
- Linear Regression and Ridge Regression produced nearly identical results, indicating that overfitting from large coefficients was not a significant issue in the baseline model.
- Decision Tree Regressor captured non-linear relationships in the data that the linear models could not, resulting in the lowest MAE and RMSE and the highest R^2 Score.
- The Actual vs Predicted plot confirmed stronger performance in the mid-price range, with reduced reliability at the extremes.

Conclusion

Decision Tree Regressor outperformed Linear Regression and Ridge Regression with an R^2 score of 0.600, the lowest MAE (0.522), and the lowest RMSE (0.724). Therefore, it was selected as the best-performing model for this task. This exercise demonstrated a complete, industry-aligned Machine Learning workflow — from data preprocessing and feature scaling to multi-model comparison and evidence-based model selection.